

	Classical Electron Theory (Lorentz-Drude)	Quantum Free Electron Theory (Sommerfeld-Drude)	Band Theory of materials
Explanation for conductivity	Newtonian motion of free electrons inside conductors	Incorporates Quantum Mechanics and Fermi-Dirac statistics	The discrete energy levels split into closely spaced energy states which form continuous energy bands, such as valence and conduction band
Basic assumptions	- Metals consist of +ve ion cores and valence e-s - Valence e-s are free and move at the order of 10^6 m/s - e-s obey classical Newtonian mechanics - e-s follow Maxwell-Boltzmann distribution of velocities, behaving like gas molecules in a container	- e-s are treated as waves obeying schrodinger equation - Potential experienced by electrons is zero inside the metal and infinite at the boundaries (simplifies complex lattice interactions) - e-s have discrete energy levels due to this potential - e-s are distributed into these energy levels as per Fermi-Dirac distribution function	- e-s are treated as waves obeying schrodinger equation - Assumes that potential energy of electrons inside the lattice is minimum at +ve ion sites and maximum between two ions. These potential energy values repeat with the periodicity of the crystal of the material.
Successes	Provides basic expressions for drift velocity and conductivity	- Gives better values for electrical conductivity - Temperature dependence of conductivity is better explained - Electronic contribution to specific heat is very small, agreeing with experimental results	- Explains semiconductors and insulators - Accounts for periodic potential of +ve ions - Accounts for +ve Hall coefficients
Drawbacks	- Cannot explain classification of materials - Predicts that multivalent metals always have higher conductivity than monovalent metals - Predicts that conductivity decreases linearly with temperature - Predicts that e-nic contribution to specific heat is comparable to that of lattice vibrations - Cannot account for positive Hall coefficients (+ve charge carriers)	- Fails for semiconductors and insulators - Cannot account for positive Hall coefficients - Ignores periodic potential of positive ions in the lattice	-